

The forbidden-branch sieve, across the collection

WS3 transfer sweep — machine 3's halting branch is pinned to a single valuation, and the sieve's reach is governed by the thinness of the halting set — July 26, 2026

1. The sieve in machine-independent form

WS3's exclusion was derived for the Space Needle from that machine's own algebra. Stripped to its skeleton it needs only that the return map be affine on each branch. Let F act on branch b as $F(x) = \alpha_b x + \beta_b$ with $\alpha_b = N_b/D_b$ in lowest terms and fixed point $x_b^* = P_b/Q_b$, also in lowest terms. A single step on branch b satisfies $D_b(Q_b x_1 - P_b) = N_b(Q_b x_0 - P_b)$, and since $\gcd(N_b, D_b) = 1$ this forces

$$N_b \text{ divides } Q_b x_1 - P_b \quad (*)$$

If the step ends in a halt then x_1 lies in the halting set H , so a halt can follow a b -step only if some h in H satisfies $Q_b h \equiv P_b \pmod{N_b}$. When that has no solution the branch is **forbidden**: no orbit, from any start, at any scale, at any time, can halt out of it. No run is needed — a single step already carries the content, and iterating $(*)$ n times gives the run version used for the Needle.

Each machine supplies only its branch data (N, D, P, Q) and a decision procedure for membership in H . Everything else is shared code.

2. Machine 3: the halting branch is a single valuation

Machine 3's reset map, verified against its own accelerated step, is $G(a) = (3^{j+1}+1)m + (r \cdot 3^j + j + c_r)$ on the branch $a = 3^j(3m + r)$, r in $\{1,2\}$, $c_1 = 3$, $c_2 = 4$. In affine form $N = 3^{j+1}+1$, $D = 3^{j+1}$, and the fixed point is an *integer*, $x^* = r \cdot 3^j - 3^{j+1}(j + c_r)$, so $Q = 1$. The halting set is the powers of 27. The sieve condition is therefore

$$27^k \equiv r \cdot 3^j + j + c_r \pmod{3^{j+1} + 1}$$

using $3^{j+1} \equiv -1$. Exact discrete logarithm says every branch with $j \leq 20$ fails except (1,1) and (1,2). The reason is visible and gives a proof for *all* j . Because $3^{j+1} \equiv -1$, the group generated by 3 modulo N is exactly $\{3^i\}$ together with $\{N - 3^i\}$ for $i \leq j$ — only $2(j+1)$ residues, and every power of 27 is one of them. Sorted, those residues leave a **large gap**: nothing lies strictly between 3^j and $2 \cdot 3^j + 1$, and nothing between $2 \cdot 3^j + 1$ and $2 \cdot 3^j + 2 \cdot 3^{j-1} + 1$. For $j \geq 2$ the fixed point lands strictly inside one of those gaps: $r = 1$ gives $3^j + j + 3$ in the first (since $j+2 < 3^j$), $r = 2$ gives $2 \cdot 3^j + j + 4$ in the second (since $j+3 < 2 \cdot 3^{j-1}$). And $j = 0$ is excluded because $N = 4$ there while every power of 27 is odd.

Theorem (machine 3's halting branch). Machine 3 can halt only out of a step whose branch has $v_3(a) = 1$. Unconditional, every start, every scale, every time.

This is qualitatively stronger than the Needle result. There the excluded valuations were a scattered 19 of the first 35, leaving an infinite allowed tail; here the allowed set is a **single valuation**. Since $P(v_3 = 1) = 2/9$, it removes $7/9 = 77.78\%$ of all steps — measured 77.61% over the orbit's first 100,000 steps.

3. The sieve iterates: the last two steps are both pinned

Because H is geometric and the branches are affine, the preimage of a geometric family is again a geometric family, so the sieve can be run backwards. Solving (*) exactly gives the depth-1 halting seeds in closed form — two families, with $R = 27^4$:

$$F_1 = \{ (3^5 R^t - 33)/10 \} = \{21, 12914013, \dots\} \quad F_2 = \{ (3^{14} R^t - 39)/10 \} = \{478293, \dots\}$$

Sieving these in turn pins the next-to-last step as well. A closed-form **parity lemma** does half of it outright: every member of F_1 and F_2 is odd and $N = 3^{j+1} + 1$ is always even, so (*) forces P odd; P has parity $r + j + c_r$ and $r + c_r$ is even for both r, hence **j must be odd** — which alone kills $j = 0$, the branch carrying two thirds of all steps. The remaining odd j are cleared by a fast exact test: multiplying (*) by 10 gives $3^{e+12t} \equiv 10P + C \pmod{N}$, whose left side ranges over at most $2(j+1)$ known residues, so membership is a tiny finite check. No j other than 1 survives, for $j \leq 500$.

Result (step –2, machine-verified). The next-to-last branch before a halt also has $v_3 = 1$. So the last two steps before any halt both have valuation 1 — excluding $1 - (2/9)^2 = \mathbf{95.06\%}$ of consecutive step pairs, measured 95.07% on the orbit.

And it stops there. Continuing the backward family enumeration shows the pinning has depth exactly 2: three steps back the valuation may be 0 or 1, four steps back 0, 1 or 2, five steps back 0 to 4. The family count grows 2, 4, 6, 17, 80 with depth, so the backward tree does not die and no decision follows. The constraint is a sharp local condition on how a halt must be approached, not a global obstruction.

step before halt	possible v_3	status
–1	{1}	PROVED for all j (gap argument), verified $j \leq 400$
–2	{1}	Parity lemma proved for all even j; verified $j \leq 500$
–3	{0, 1}	backward family enumeration, $j \leq 24$
–4	{0, 1, 2}	backward family enumeration, $j \leq 24$
–5	{0, 1, 2, 3, 4}	backward family enumeration, $j \leq 24$

4. The sweep: where the sieve bites and where it cannot

machine	halting set H	shape of H	what the sieve gives
Space Needle	$\{2^m\}$	geometric	19 of the first 35 valuations forbidden; 28.7% of steps
Machine 3	$\{27^k\}$	geometric	v_3 pinned to 1 at each of the last two steps; 77.78% of steps, 95.06% of step pairs
Fenrir	$\{n : 4 \text{ divides } n\}$	a congruence class	vacuous — $\gcd(4, 5) = 1$, so both branches are solvable and nothing is excluded
Hydra / Antihydra	first passage of a walk	not a set of values	vacuous — halting is not a condition on the iterate's value, so (*) says nothing
Machine 4	$b = a + 3, a \text{ odd}$	a line in two variables	out of scope as stated: no one-variable return map, so there is no single branch-affine coordinate to sieve

The pattern. The sieve's strength is governed by how thin H is as a *set of values*, and nothing else. A geometric H makes (*) a discrete-logarithm condition that can fail, and the sparser the group generated by the base modulo N , the more often it does — which is why base 3 beats base 2 so decisively here: modulo $3^{j+1}+1$ the powers of 3 collapse to $2(j+1)$ residues with huge gaps between them, whereas modulo $2^{v+1}+3$ the powers of 2 are spread out. A congruence-class H is immune, because a congruence class meets every residue class of any coprime modulus. And a halting condition that is not about the iterate's value at all — a first passage, a coincidence between two counters — is outside the method entirely. This also says where to look next: the sieve is a tool for multiplicative-coincidence cryptids, and only for those.

5. Verification ledger

claim	status
The general form (*)	PROVED (affine conjugation to the fixed point; $\gcd(N, D) = 1$). The derived branch data and (*) were machine-checked against machine 3's own verified map on 39,989 values $a < 40,000$.
Machine 3: $v_3 = 1$ at the halting step	PROVED for all j by the gap argument, with each step of the proof machine-checked (the description of the group generated by 3, the two gaps, and the inequalities placing the fixed point inside one) for $j \leq 400$; and cross-checked by direct membership test, no gap argument used, for $j \leq 59$.
Machine 3: $v_3 = 1$ at the next-to-last step	PARTLY PROVED. The parity lemma (j odd) is closed form for all j . The elimination of the remaining odd j is MACHINE-VERIFIED for $j \leq 500$ by an exact finite test, not proved in general.
Independent confirmation	MACHINE-VERIFIED by brute force with no sieve theory used: over all $a < 4,000,000$, every value halting within three steps has $v_3 = 1$ at the last step and at the next-to-last step. The only two-step seed found is $15 \rightarrow 21 \rightarrow 27$.
Orbit measurements	MACHINE-VERIFIED over 100,000 steps of the true reset orbit: 77.61% of steps and 95.07% of consecutive pairs excluded, against asymptotic 77.78% and 95.06%. No halt; the orbit never touched the spine.
Depth of the pinning	MACHINE-VERIFIED for branches $j \leq 24$ at depths up to 5. The loosening at step -3 is a positive finding (explicit surviving families exist), not a failure to exclude.
Non-halting	OPEN for every machine here. The sieve constrains how a halt must be approached; it excludes no orbit.

Code: sieve.py (the general sieve), sieve_m3.py (machine 3 branch data and brute-force check), m3_theorem.py (the gap proof), m3_step2.py (parity lemma and the large- j test), m3_deep.py (backward family enumeration).